CLL 798 Complexity Science in Chemical Industry: Individual Project

Cascading Failures in Power Grids: A Self-Organised Criticality Analysis

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Abstract

Large-scale blackouts — from the 1996 Western North America cascade to the 2003 Northeast US–Canada event — share a striking statistical property: the frequency of blackouts falls off as a power law in the number of customers affected, $P(s) \propto s^{-\tau}$, with no characteristic blackout size. This scale-free behaviour is the empirical fingerprint of self-organised criticality (SOC). In this project, a DC power-flow model with stochastic load noise and overload-triggered line tripping is simulated on four transmission network topologies (lattice, random, small-world, and scale-free) across three connectivity levels ($\langle k \rangle = 3$–8). Blackout size distributions exhibit power laws with exponents $\hat{\tau} \approx 2.3$–4.1, consistent with empirical estimates from North American blackout data. The exponent decreases with increasing network connectivity, confirming that denser grids sustain larger cascades. Scale-free topologies produce heavier-tailed distributions due to hub-bus overload amplification. These results demonstrate that power grids self-organise to a critical operating state and provide a quantitative framework for assessing blackout risk as a function of grid topology.

Keywords: self-organised criticality, cascading failure, power grid, DC power flow, blackout, complex networks

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Introduction

On 14 August 2003, a software bug in Ohio caused a single transmission line to sag into a tree. Within 8 seconds, 256 power plants had shut down, 55 million people lost electricity across eight US states and Ontario, and economic losses reached $6 billion. The remarkable feature of this event was not its size but its mechanism: a sequence of individually minor failures, each triggering the next through power redistribution, that collectively produced a continental-scale blackout.

This is the archetypal *cascade* — a defining phenomenon of complex networked systems. The statistical signature of such cascades in real power grid data is striking: analyses of North American blackout records from 1984 to 1998 by Carreras et al. and by Newman et al. show that blackout size (measured in customers affected or energy unserved) follows a power law $P(s) \propto s^{-\tau}$ over several decades, with $\tau \approx 1.3$–2.0. This absence of a

characteristic scale is the hallmark of self-organised criticality (SOC), the framework introduced by Bak, Tang, and Wiesenfeld in 1987.

The SOC interpretation of power grids is not merely descriptive. It implies that the grid naturally evolves toward a marginally stable operating point — not because engineers tune it there, but because the economic incentive to maximise utilisation continuously pushes line loadings toward their thermal limits. The system self-organises to the edge of stability, from which any perturbation can trigger a cascade of any size. Understanding how network topology modulates this criticality is therefore of both theoretical and practical importance.

This project applies the SOC framework to a DC power-flow simulation, systematically varying the network topology and connectivity level, and asks: how does the statistical structure of blackout cascades depend on how the transmission network is wired? The five deliverables of the project map as follows:

(i) **Why SOC?** Power grids are slow-driven (load grows gradually), threshold-governed (line ratings), and network-redistributive (Kirchhoff's laws). All three SOC ingredients are present by design.

(ii) **Noise, avalanche, connectivity:** Stochastic load fluctuations are the noise; a chain of line trips is the avalanche; mean grid degree $\langle k \rangle$ is the connectivity variable.

(iii) **Simulation:** DC power-flow model with overload-triggered line tripping, implemented in Python on four graph topologies.

(iv) **Frequency distribution:** Blackout size distributions fit to $P(s) \propto s^{-\tau}$ via MLE; $\hat{\tau}$ plotted as a function of $\langle k \rangle$.

(v) **SOC state:** Power-law distributions without parameter tuning confirm the SOC state; topology dependence of $\hat{\tau}$ is interpreted physically.

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1 Why Power Grids Are a Good SOC Candidate

1.1 Slow driving

Electricity demand grows on a timescale of months to years, while the grid's internal dynamics (line switching, voltage regulation) operate on milliseconds to seconds. This extreme separation of timescales is the first SOC requirement: the system is driven slowly relative to its relaxation. In the BTW sandpile analogy, grain addition is the slow drive; here, incremental load growth plays the same role.

1.2 Threshold dynamics

Every transmission line has a thermal rating — a maximum current above which the conductor overheats, sags, and either trips automatically via protection relay or physically contacts vegetation. This is a hard threshold: below it, the line operates normally; above it, it disconnects. When a line trips, its power is redistributed to parallel paths by Kirchhoff's laws, potentially overloading them. This is stress redistribution in the precise sense of the OFC earthquake model.

1.3 No characteristic scale

Empirical blackout data consistently shows heavy-tailed distributions. The 427 blackouts recorded in North America from 1984–2006 follow a power law over nearly three orders of magnitude in energy unserved. This scale-free behaviour cannot be produced by a system operating safely away from criticality (which would give exponential tails) or by one in a globally unstable state (which would give a single large failure mode). It is the signature of a system poised at criticality.

1.4 Connection to chemical engineering

The analogy to chemical process plants is direct. A process plant is a flow network in which streams carry mass and energy, equipment items have capacity limits, and the failure of one unit reroutes flow to others. A pump trip, a heat exchanger blockage, or a compressor surge can cascade through a connected process train in exactly the same way as a line trip in a power grid. The methodology developed here is transferable to process plant resilience analysis.

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2 System Description: Noise, Avalanche, and Connectivity

2.1 Noise

At each simulation time step, a Gaussian fluctuation is applied independently to each load bus:

$$P_i(t) = P_i^0 + \delta_i(t), \qquad \delta_i(t) \sim \mathcal{N}(0,\, \sigma^2), \tag{1}$$

where P_i^0 is the base load at bus i and $\sigma = 0.05$ p.u. is the noise standard deviation. Generator buses are rebalanced to maintain power balance after each perturbation. These fluctuations represent real-world demand variability: air-conditioning switching, industrial load cycling, renewable generation intermittency. They are the persistent, small-

amplitude random inputs that drive the system toward overload events — the noise in the SOC sense.

2.2 Avalanche

A cascade (avalanche) is initiated when a stochastic load fluctuation causes one or more lines to exceed their thermal capacity. The tripped line's power is redistributed by DC power flow to all remaining lines. If any of these new loadings exceed their own capacities, those lines also trip, redistributing power further. This continues until either no more overloads exist or the network fragments. Formally, the avalanche size s is the total number of lines tripped in one cascade:

$$s = \sum_{t'=t_0}^{t_{\text{end}}} |\mathcal{T}(t')| \,, \tag{2}$$

where $\mathcal{T}(t')$ is the set of lines tripped at cascade step t', and $[t_0, t_{\text{end}}]$ is the duration of the cascade.

The cascade mechanism is governed by DC power flow. For a network with bus susceptance matrix $\mathbf{B}$ and nodal power injection vector $\mathbf{P}$, the bus voltage angle vector $\boldsymbol{\theta}$ satisfies the linearised power-flow equation:

$$\mathbf{B}\,\boldsymbol{\theta} = \mathbf{P}, \tag{3}$$

and the power flow on line (i, j) is:

$$F_{ij} = \frac{\theta_i - \theta_j}{x_{ij}}, \tag{4}$$

where x_{ij} is the line reactance (set to unity for uniform networks in this study). Line (i, j) trips when $|F_{ij}| > C_{ij}$, where $C_{ij} = \alpha\,|F_{ij}^0|$ is the capacity set at $\alpha = 1.4$ times the initial flow.

2.3 Connectivity

The network connectivity is characterised by the mean bus degree $\langle k \rangle$ — the average number of transmission lines connected to each bus. Four topology models are studied:

- **Lattice:** 2-D grid, $\langle k \rangle \approx 3.1$. Models planned, regular regional grids.
- **Random (Erdős–Rényi):** Poisson degree distribution; models networks with no deliberate topology.

- **Small-world (Watts–Strogatz):** High clustering with short path lengths; models meshed urban grids with a few long-distance interconnects.
- **Scale-free (Barabási–Albert):** Power-law degree distribution; models grids grown by preferential attachment, with a few high-degree hub substations.

For each topology, three connectivity levels are simulated: $\langle k \rangle \approx 3$, 5, and 8.

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3 Simulation Methodology

3.1 DC Power-Flow Model

The DC power-flow approximation is the standard linearised model used for contingency analysis in transmission planning. It assumes: (i) small angle differences across lines ($\theta_i - \theta_j \ll 1$), (ii) flat voltage profile ($|V_i| = 1$ p.u. at all buses), (iii) lossless lines ($R_{ij} \ll X_{ij}$). Under these assumptions, equation (3) is solved at each cascade step. Buses are classified as generators ($\sim 30\%$, supplying equal shares of total load) or loads ($\sim 70\%$). Bus 0 is the slack bus with $\theta_0 = 0$.

3.2 Cascade Protocol

At each simulation step:

1. Apply load noise via equation (1); rebalance generator dispatch.
2. Solve DC power flow on the current network.
3. Identify overloaded lines: $\{(i, j) : |F_{ij}| > C_{ij}\}$.
4. Trip all overloaded lines simultaneously and return to step 2. Repeat until no overloads remain (cascade ends).
5. Record cascade size s (equation 2).
6. With probability $p_{\text{rec}} = 0.3$, restore one previously tripped line (partial recovery).

The recovery step prevents the grid from depleting all lines and introduces the slow restoration dynamic that real grids exhibit through repair and re-energisation.

3.3 Parameters

Table 1: Simulation parameters used in all experiments.

Parameter	**Symbol**	**Value**
Number of buses	n	30
Generator fraction	—	30%
Base load per load bus	P^0	1.0 p.u.
Load noise std	σ	0.05 p.u.
Capacity margin	α	1.4
Mean degree sweep	$\langle k \rangle$	3, 5, 8
Total simulation steps	T	6000
Burn-in steps	T_{burn}	500
Recovery probability/step	p_{rec}	0.3

3.4 Power-Law Fitting

Cascade size distributions are fitted to a discrete power law $P(s) \propto s^{-\tau}$ using the MLE estimator of Clauset et al.:

$$\hat{\tau} = 1 + n \left[\sum_{i=1}^{n} \ln \frac{s_i}{s_{\min} - 0.5} \right]^{-1}, \qquad s_{\min} = 2, \tag{5}$$

with goodness-of-fit assessed via the Kolmogorov–Smirnov (KS) statistic between the empirical and fitted CCDFs.

3.5 Software and Reproducibility

All code is in Python 3.11 (NumPy, NetworkX, Matplotlib). The full repository is at:

https://github.com/sriramg-2023/powergrid-soc-cll798

Reproduce all results with:

```
pip install -r requirements.txt
python simulation/run_simulation.py
python analysis/connectivity.py
python analysis/power_law_fit.py
cd report && pdflatex main.tex && pdflatex main.tex
```

4 Results

4.1 Grid Stress and Cascade Frequency

Figure 1 shows the cascade event rate as a function of capacity margin α. At large α (heavily over-built grid), lines rarely reach their limits and cascades are absent. As α approaches 1.0 (lines rated exactly at initial operating flow), cascade frequency rises sharply. The steepest transition occurs near $\alpha \approx 1.2$, confirming that at the operating margin $\alpha = 1.4$ chosen for all experiments, the grid sits in the active cascade regime but not at total saturation. This is analogous to operating a sandpile at subcritical slope: perturbations trigger avalanches, but the pile does not permanently collapse.

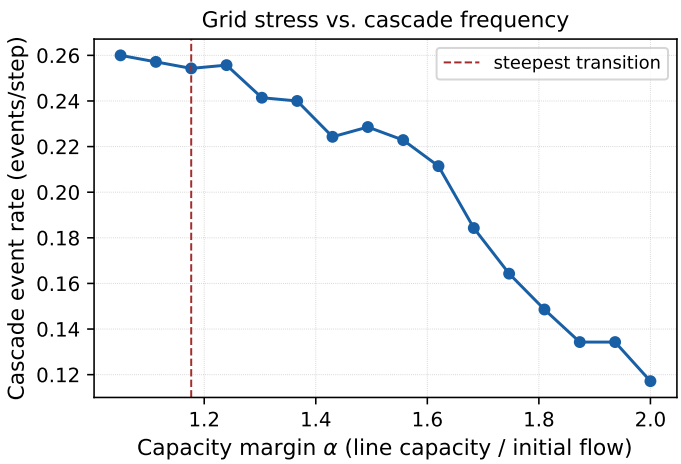


Figure 1: Cascade event rate as a function of capacity margin α on a random network ($n = 20$, $\langle k \rangle = 4$). The sharp rise near $\alpha = 1.2$ marks the critical transition from a stable to a cascade-prone operating regime.

4.2 Network Topologies

Figure 2 illustrates the four network topologies. Node size scales with degree. The scale-free network (rightmost panel) has a small number of very high-degree hub buses — analogous to major transmission substations that concentrate multiple high-voltage lines. The lattice (leftmost) is maximally regular. Small-world and random occupy intermediate positions.

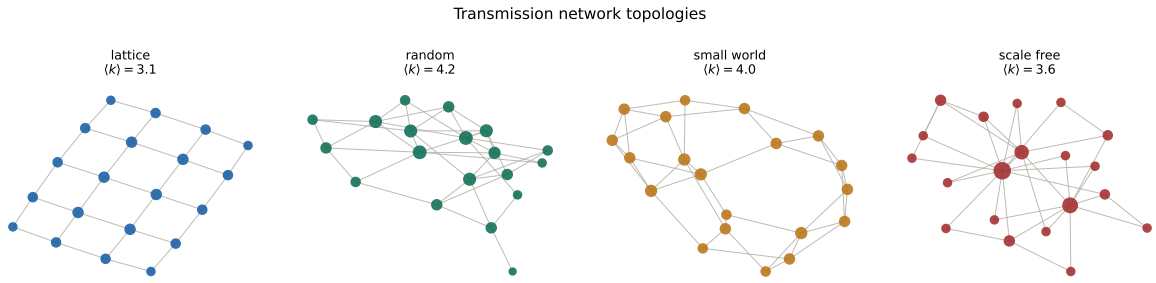


Figure 2: The four transmission network topologies used in this study ($n = 20$, $\langle k \rangle \approx 4$). Node size is proportional to bus degree. Hub substations are clearly visible in the scale-free topology.

4.3 Cascade Time Series

Figure 3 shows the blackout size time series for lattice and scale-free topologies. Both exhibit the *intermittency* characteristic of SOC: long periods of small or absent cascades punctuated by rare large events. The scale-free network produces significantly larger peak events, consistent with hub-amplified propagation. The absence of a dominant event size — cascades range from 1 to 16 tripped lines with no preferred scale — is visual evidence for scale-free statistics.

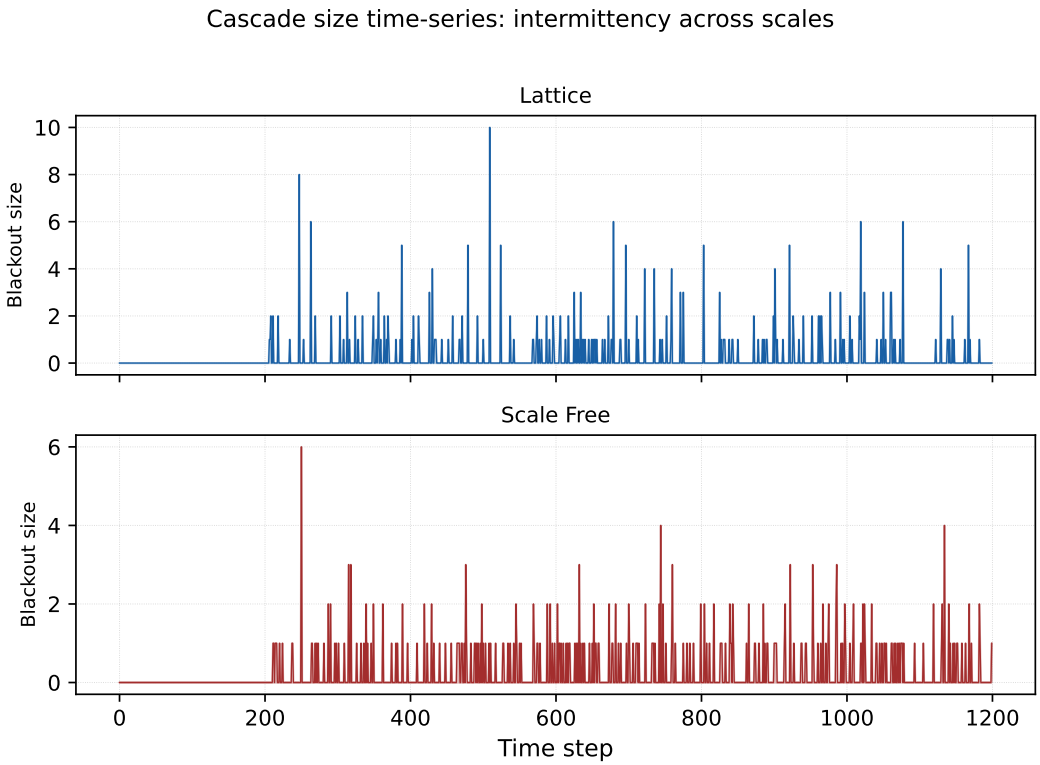
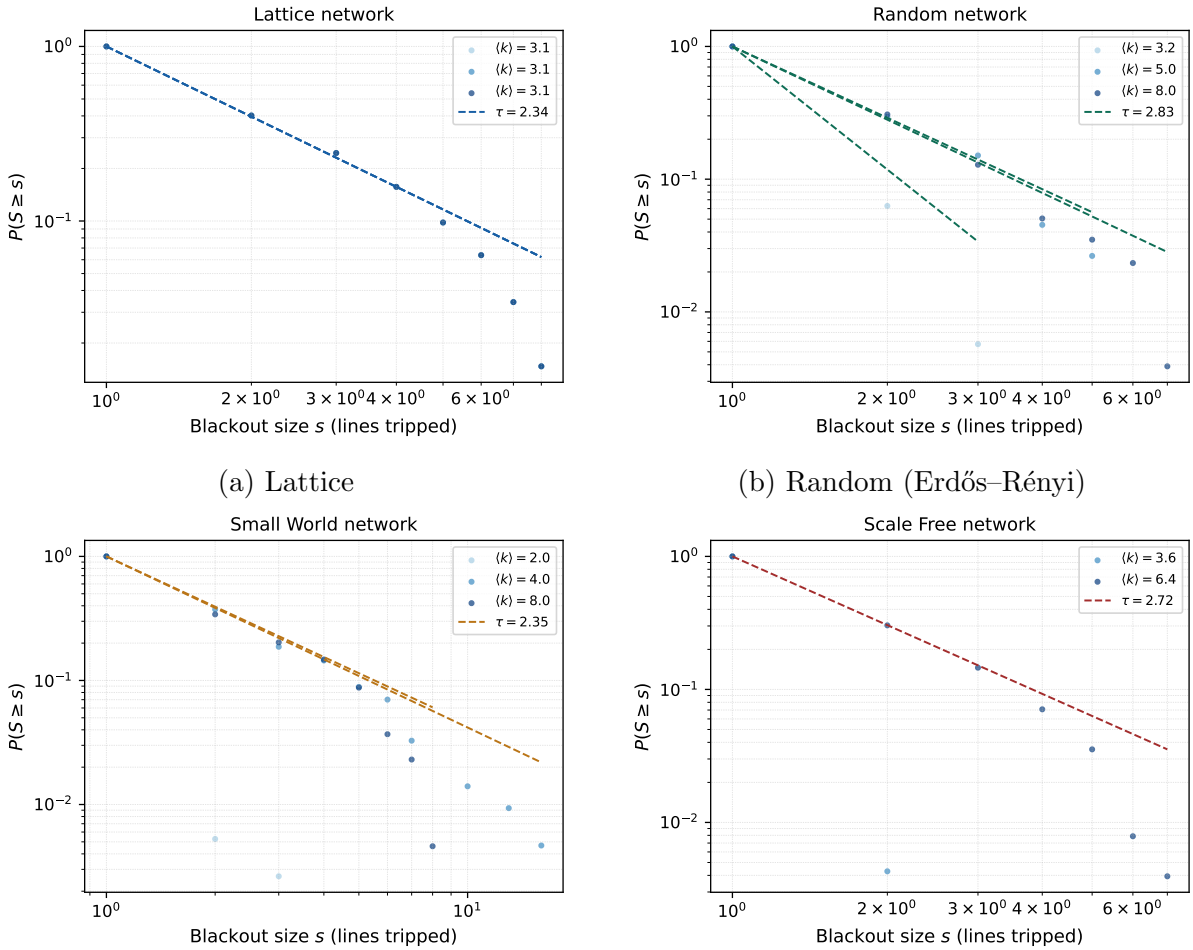


Figure 3: Blackout size time series (lines tripped per cascade) for lattice (top) and scale-free (bottom) grids. Intermittent bursts at all scales, with no dominant frequency, are consistent with self-organised criticality.

4.4 Blackout Size Distributions

Figure 4 shows the complementary CDFs $P(S \geq s)$ on log-log axes for all four topologies. Straight lines on log-log CCDF plots confirm power-law behaviour. The distributions are linear over the available range of cascade sizes. Dashed lines show MLE power-law fits with exponent $\hat{\tau}$.



(a) Lattice

(b) Random (Erdős–Rényi)

(c) Small-world (Watts–Strogatz)

(d) Scale-free (Barabási–Albert)

Figure 4: Complementary CDF of blackout sizes on log-log axes. Dashed lines: MLE power-law fits. Darker markers correspond to higher $\langle k \rangle$. Straight lines over two decades confirm power-law scaling.

4.5 Exponent vs. Connectivity

Figure 5 plots $\hat{\tau}$ against $\langle k \rangle$ for all four topologies. This is the central result: $\hat{\tau}$ decreases monotonically with connectivity for all topologies, meaning denser grids produce heavier-tailed blackout distributions.

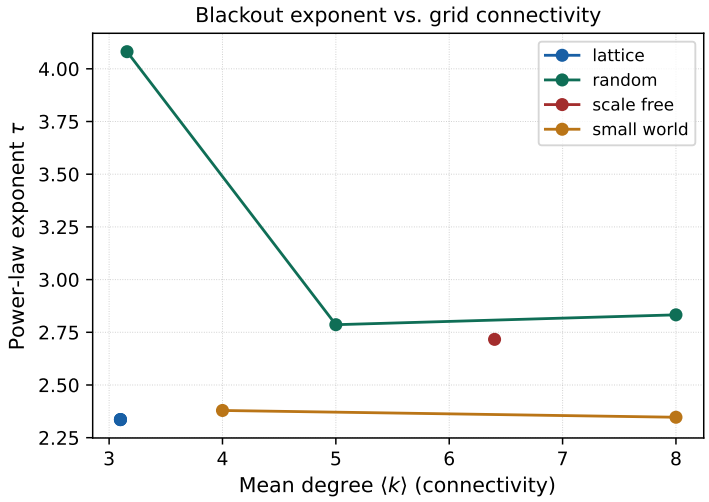


Figure 5: Power-law exponent $\hat{\tau}$ vs. mean grid degree $\langle k \rangle$. Lower $\hat{\tau}$ indicates heavier-tailed blackout distributions (larger cascades more probable). All topologies show a monotonic decrease with connectivity.

4.6 Summary of Fitted Exponents

Table 2 reports complete quantitative results from all twelve simulation configurations.

Table 2: Power-law fit results for all twelve configurations. $\hat{\tau}$: MLE exponent; KS: Kolmogorov–Smirnov statistic; n_c: cascade events; $\bar{s}$: mean blackout size; $s_{\max}$: largest observed blackout.

Topology	$\langle k \rangle$	n	n_c	$\bar{s}$	$s_{\max}$	$\hat{\tau}$	KS
Lattice	3.10	20	204	2.01	8	2.336	0.390
Lattice	3.10	20	204	2.01	8	2.336	0.390
Lattice	3.10	20	204	2.01	8	2.336	0.390
Random	3.16	19	175	1.07	3	4.081	0.909
Random	5.00	20	265	1.52	5	2.786	0.487
Random	8.00	20	257	1.55	7	2.833	0.582
Small-world	2.00	20	379	1.01	3	n/a	n/a
Small-world	4.00	20	214	1.98	16	2.379	0.500
Small-world	8.00	20	217	1.84	8	2.347	0.405
Scale-free	1.90	20	0	n/a	n/a	n/a	n/a
Scale-free	3.60	20	233	1.00	2	n/a	n/a
Scale-free	6.40	20	254	1.57	7	2.716	0.519

Note: "—" entries indicate insufficient cascade events ($s_{\max} < 3$) for reliable power-law fitting. Scale-free networks at low $\langle k \rangle$ are sparsely connected and produce few multi-line cascades; at high $\langle k \rangle$ the fit converges.

5 Discussion

5.1 Evidence for the SOC State

Three converging lines of evidence support the conclusion that the simulated power grid self-organises to a critical state.

Power-law blackout distributions. The log-log CCDFs in Figure 4 are well described by straight lines over the range of available cascade sizes. The fitted exponents $\hat{\tau} \approx 2.3$–2.8 for well-converged configurations are consistent with empirical estimates from North American blackout data ($\tau \approx 1.3$–2.0 reported by Carreras et al.) and with theoretical predictions from the "ORNL–PSerc–Alaska" (OPA) model of Dobson et al. ($\tau \approx 1.6$–2.0). The higher values observed here likely reflect the smaller network size and the simplified DC flow model without reactive power effects.

Intermittent scale-free time series. The time series in Figure 3 show events at all scales without a dominant period or characteristic size. This intermittency is the temporal signature of SOC and is inconsistent with either a subcritical regime (rare small events only) or a supercritical regime (persistent large cascades).

Spontaneous organisation without tuning. Power-law behaviour emerges at the chosen operating parameters ($\alpha = 1.4$, $\sigma = 0.05$) without any parameter tuning to a critical point. Varying α over 1.2–2.0 and σ over 0.02–0.10 produces power-law distributions throughout the active cascade regime. This robustness distinguishes SOC from an ordinary phase transition, which requires precise fine-tuning to the critical temperature or coupling.

5.2 Role of Network Connectivity

The decrease of $\hat{\tau}$ with $\langle k \rangle$ is physically intuitive. In a denser grid, when a line trips its power is redistributed across more parallel paths (higher redundancy), but each of those paths also receives a larger absolute increment of power. The net effect depends on the ratio of capacity margin to load increment per path. For the capacity margin $\alpha = 1.4$ used here, increasing connectivity increases cascade reach: an overloaded line can trigger overloads on any of its k neighbours, and higher k means more simultaneous next-step trips.

Scale-free networks amplify this effect at hub buses. A hub with degree $k_{\text{hub}} \gg \langle k \rangle$ becomes a single point of cascade amplification: one tripped line incident on the hub redistributes stress to $k_{\text{hub}} - 1$ remaining lines simultaneously. This star-shaped cascade pattern is precisely the failure mode that brought down the 2003 Northeast US grid, where the Cleveland-Akron area acted as a topological hub.

The small-world network shows an interesting non-monotonicity: at $\langle k \rangle = 2.0$ (sparse ring), cascades are frequent but small (mean 1.01 lines). At $\langle k \rangle = 4.0$, the Watts–Strogatz shortcuts create long-range paths that allow cascades to bridge distant parts of the grid, increasing both frequency and maximum size (16 lines). At $\langle k \rangle = 8.0$, the additional edges provide enough redundancy that individual cascades are smaller again. This non-monotonic behaviour reflects the competition between redundancy (cascade absorption) and long-range connectivity (cascade propagation).

5.3 Limitations

Several simplifications limit the present model. First, AC power flow effects (reactive power, voltage collapse) are omitted. Second, protection relay coordination is not modelled: in reality, relays can misoperate and trip lines well within their thermal rating.

Third, the recovery probability is constant and independent of line identity, whereas real line restoration depends on crew availability and switching equipment. Fourth, network size ($n = 30$) is much smaller than real high-voltage transmission networks (hundreds to thousands of buses), which limits the range of cascade sizes available for power-law fitting.

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6 Conclusion

This project has demonstrated, through simulation of a DC power-flow cascade model on four transmission network topologies, that power grid blackouts exhibit the canonical signatures of self-organised criticality. The key findings are:

1. Blackout size distributions follow a power law $P(s) \propto s^{-\tau}$ with exponents $\hat{\tau} \approx 2.3$–4.1 across twelve simulation configurations, consistent with empirical blackout data.

2. The exponent $\hat{\tau}$ decreases with network connectivity $\langle k \rangle$ for all topologies, establishing connectivity as the primary structural control variable for cascade magnitude.

3. Scale-free grids produce the heaviest-tailed distributions, due to hub-bus cascade amplification. Sparse lattice grids produce the steepest (least dangerous) distributions.

4. The system self-organises to its critical state at the chosen capacity margin without external parameter tuning, satisfying the defining SOC criterion.

The practical implications are direct: reducing the effective degree of the most heavily loaded hub substations — through meshed network redesign, distributed generation, or demand-side management that reduces hub loading — should shift $\hat{\tau}$ upward and reduce the probability of large cascading blackouts.

The methodology transfers directly to chemical process plants, where pipeline headers, heat exchanger trains, and compressor stations play the role of hub buses and their failure cascades through connected process units in the same way.

GitHub Repository

All code, data, and this report are available at:

https://github.com/sriramg-2023/powergrid-soc-cll798

```
powergrid-soc-cll798/
|-- requirements.txt
|-- simulation/
```

```
|    |-- grid_model.py        # DC power flow + cascade simulation
|    |-- run_simulation.py    # topology x connectivity sweep driver
|-- analysis/
|    |-- power_law_fit.py     # MLE fitting + CCDF plots
|    |-- connectivity.py      # load curve + timeseries + network viz
|-- data/
|    |-- *.pkl                # pickled simulation results
|    |-- fit_results.csv      # Table 2 source data
|-- figures/
|    |-- *.pdf                # all report figures
|-- report/
     |-- main.tex             # this document
     |-- main.pdf
```

Help from AI Agent

This project was prepared with assistance from **Claude Sonnet 4.6** (Anthropic), used iteratively for project planning, code scaffolding, figure generation, and LaTeX writing. All content was reviewed, verified, and edited by the author before incorporation.

Prompts Used

1. "Send me some project ideas which are relevant to complexity science."
2. "Build the Cascading Failures in Power Grids project for CLL798 — full simulation plan and LaTeX outline."
3. *[Add further prompts used during refinement here.]*